Phase 1 of 8: Core language foundations · lesson 1 of 20 · checklist · book reading

C++ Basics and Pointers

Start here if C++ is new or rusty: types, functions, where variables live, and how pointers and references work.

How a C++ program is built



Compile error: a problem inside one file. Linker error: declared and called but never defined.

Three stages. Know which one an error comes from.

C++ is **compiled** to machine code before it runs: fast, but it trusts you not to make memory mistakes.

main.cpp

```cpp
#include <iostream>          // paste in the declarations for std::cout

int main() {                 // execution starts here
    std::cout << "Hello, C++\n";
    return 0;                // 0 means success
}
```

1. **Preprocess**: `#include` literally pastes header files in; `#define` does text substitution.
2. **Compile**: each `.cpp` file becomes an *object file* on its own.
3. **Link**: the linker joins object files and libraries into one executable and resolves calls between files.

> HABIT TO BUILD NOW Always compile with `-Wall -Wextra`. The compiler warns about many of the bugs this lesson describes (uninitialized variables, unused values, signed/unsigned comparisons).

KEY POINT "Compile error vs linker error?" A compile error is a syntax or type problem inside one file. A linker error ("undefined reference") means a function was declared and called but never defined anywhere you linked.

Variables and types

Every variable has a **type** that fixes how much memory it uses and what you may do with it.

TYPE	HOLDS	TYPICAL SIZE
`bool`	`true` / `false`	1 byte
`char`	a character / small integer	1 byte (always, by definition)
`int`	whole number	4 bytes
`long long`	bigger whole number	8 bytes
`float` / `double`	decimal number	4 / 8 bytes

Only `char` has a fixed size. When size matters use `<cstdint>`: `std::int32_t`, `std::uint64_t`.

Initialization: always give a variable a value

```
int a;          // BAD: indeterminate value, reading it is undefined behavior
int b = 5;      // copy initialization
int c(5);       // direct initialization
int d{5};       // brace (uniform) initialization: preferred
int e{};        // empty braces: zero-initialized, e == 0
int f{3.7};     // ERROR: braces forbid narrowing double → int
auto g = 5;     // g is int: the compiler deduces the type
const int h = 9;// cannot be changed after initialization
```

UNDEFINED BEHAVIOR An uninitialized local holds garbage. Reading it is **UB**: junk, a crash, or "works" until it does not.

The signed/unsigned trap

`unsigned` wraps instead of going negative, and a mixed comparison converts the signed side to unsigned.

Operators and control flow

```
7 / 2      // 3     integer division truncates
7 % 2      // 1     remainder
7 / 2.0    // 3.5   one double operand makes the whole thing floating point

if (x > 0 && y > 0) { }   // && and || short-circuit: right side is skipped if not needed
for (int i = 0; i < 3; ++i) { }
for (int v : vec) { }     // range-based for: visits every element
while (cond) { }
switch (n) { case 1: /* ... */ break; default: break; }   // forget `break` and it falls through
```

> CLASSIC BUG `if (x = 5)` *assigns* 5 to x and is always true. You meant `==`. The compiler warns with `-Wall`.

Functions and passing arguments

Declare before use; the header holds declarations, the .cpp the definitions.

```
int add(int a, int b);                  // declaration
int add(int a, int b) { return a + b; }// definition
double add(double a, double b);         // overload: same name, different parameter types
void greet(std::string who = "world"); // default argument
```

Three ways to pass a parameter

```
void byValue(int n)         { n++; }          // works on a COPY
void byRef(int& n)          { n++; }          // works on the ORIGINAL
void byConstRef(const std::string& s) { }     // original, read-only, no copy
```

YOU WANT TO...	USE
read a small value (int, double, char, pointer)	by value: `int n`
read a large object without copying it	by const reference: `const std::string& s`
modify the caller's variable	by reference: `int& n`
accept "no object" as a valid argument	by pointer: `Widget* w`, and test it for `nullptr`
pass a run of elements	`std::span<int>` (C++20) or `const std::vector<int>&`, not a pointer plus a count
get a result back	return it by value; big objects such as `std::vector` are moved, not copied, so no out-parameter is needed

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Pointers from scratch

A **pointer** is a variable whose value is an **address**.

SYNTAX	MEANING
`int* p`	declare `p` as "pointer to int"
`&x`	address-of: the address of `x`
`*p`	dereference: the thing `p` points to
`nullptr`	the pointer that points to nothing

Pointers and arrays

An array name **decays** to a pointer to its first element; `p + 1` moves one *element*.

```
int a[3] = {10, 20, 30};
int* p = a;            // same as &a[0]
std::cout << *p;       // 10
std::cout << *(p + 1); // 20   (p + 1 advances by sizeof(int) bytes)
std::cout << p[2];     // 30   p[i] is exactly *(p + i)
```

CAREFUL Never dereference `nullptr` and never read past the end of an array. Both are undefined behavior. C++ will not stop you.

WHAT A POINTER DOES NOT KNOW Only an address and a type: not **how many** elements, nor **who** frees them. Root of most pointer bugs.

KEY POINT "What is the difference between `NULL` and `nullptr`?" `NULL` is a macro for a null pointer constant, usually plain `0` (an integer), so `f(NULL)` can call `f(int)` instead of `f(char*)`. `nullptr` has its own type and only converts to pointers.

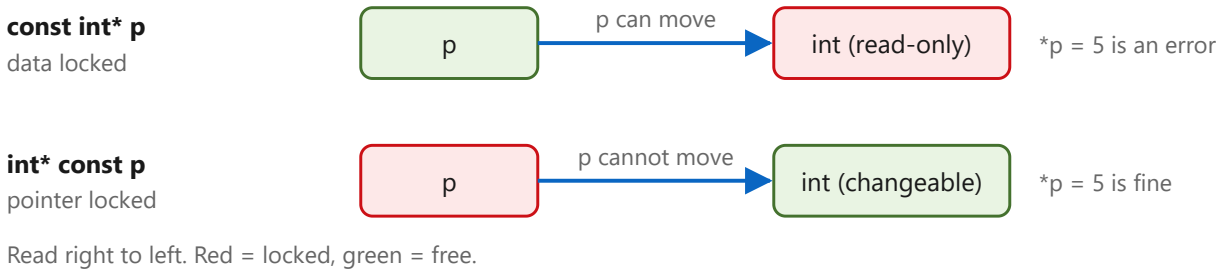
References

A reference is an **alias**: another name for an existing variable, not a copy.

```
int x = 1;
int& r = x;      // r is another name for x
r = 5;           // x is now 5
```

	POINTER `int*`	REFERENCE `int&`
Can be null	yes	no (must refer to something real)
Must be initialized	no (but should be)	yes, at declaration
Can be re-pointed	yes	no, bound for life
Syntax to use	`*p`, `p->member`	just use the name
Use for	optional things, arrays, ownership, C APIs	function parameters, aliases

`const` and pointers



Read right to left. Red = locked, green = free.

const before the star protects the data; after the star it protects the pointer.

`const` = "I will not modify this". With pointers, either the data or the pointer can be const.

Read right to left.

```
int x = 1, y = 2;

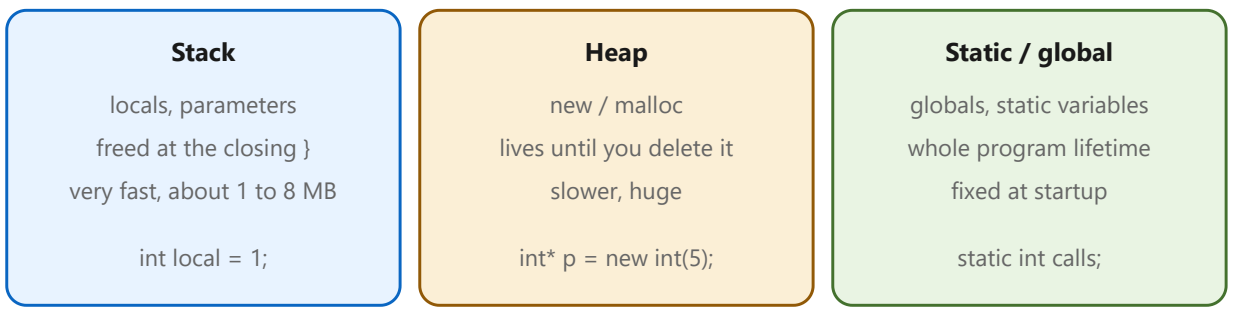
const int* p1 = &x;          // "p1 is a pointer to const int"
// *p1 = 5;                  // ERROR: cannot change the int through p1
p1 = &y;                     // OK: the pointer itself may move

int* const p2 = &x;          // "p2 is a const pointer to int"
*p2 = 5;                     // OK: the int may change
// p2 = &y;                  // ERROR: the pointer may not move

const int* const p3 = &x;    // neither can change
```

Make everything `const` that can be.

Where variables live: stack, heap, static



The lifetime rule decides where a variable should live.

REGION	WHAT GOES THERE	LIFETIME	SPEED / SIZE
Stack	local variables, function parameters	automatic: freed when the scope `{ }` ends	very fast, small (about 1 to 8 MB)
Heap (free store)	objects created with `new` / `malloc`	manual: lives until you `delete` it	slower, huge
Static / global	globals, `static` variables	whole program	fixed at startup

```
int counter = 0;                // static storage: exists for the whole program

void f() {
    int local = 1;              // stack: created on entry, destroyed on exit
    static int calls = 0;       // static storage, but only visible inside f
    ++calls;
    int* p = new int(5);        // the int lives on the HEAP; p itself is on the stack
    delete p;
}
```

> MENTAL MODEL Stack: a pile of plates, popped automatically. Heap: a storage unit you rent until you hand the key back.

Dynamic memory: `new` and `delete`

`new` allocates on the heap **and** constructs; you must `delete` it or it **leaks**.

```
int*  one = new int(42);     delete one;        // single object: new / delete
int*  arr = new int[100];    delete[] arr;      // array: new[] / delete[] (must match!)
```

BUG	WHAT HAPPENED
Memory leak	you lost the pointer (or forgot `delete`) so the memory is never freed
Dangling pointer	you used a pointer after the memory was freed
Double delete	you freed the same memory twice: undefined behavior, often a crash
Mismatch	`new[]` freed with `delete`, or the reverse

MODERN C++ Avoid raw `new` / `delete`: use `std::vector` and `std::unique_ptr` (Smart Pointers).

KEY POINT "What does `new` do beyond `malloc`?" It calls the constructor, throws `std::bad_alloc` on failure (instead of returning null), and returns a correctly typed pointer. `delete` calls the destructor and then frees the memory.

Arrays, C-strings and `std::string`

```
int a[5] = {1, 2, 3};        // fixed size, remaining elements are 0, NO
bounds checking
a[10] = 7;                   // compiles, undefined behavior at runtime

void f(int arr[]) {          // this is really `int* arr`: the size is lost!
    sizeof(arr);             // size of a POINTER (8), not of the array
}

char s[] = "hi";             // 3 chars: 'h', 'i', and a terminating '\0'
```

Prefer:

INSTEAD OF	USE	WHY
`int a[N]`	`std::array<int, N>`	knows its size, no decay, same speed
`new int[n]`	`std::vector<int>`	grows, frees itself, `.at(i)` bounds-checks
`char*` strings	`std::string`	owns its memory, safe copy, `+`, `.size()`, `.find()`
pointer plus a count	`std::span<int>` (C++20)	a non-owning view that remembers its size; accepts an array or a vector without copying

```
std::string s = "hello";
s += " world";                          // grows automatically
std::cout << s.size() << s.substr(0, 5);   // 11 hello
std::vector<int> v = {1, 2, 3};
v.push_back(4);                         // size is now 4
```

The classic memory bugs

Most "mysterious" crashes are one of these.

BUG	EXAMPLE
Uninitialized read	`int n; use(n);`
Out-of-bounds	`int a[3]; a[3] = 1;` (valid indices are 0 to 2)
Null dereference	`int* p = nullptr; *p = 1;`
Use after free	`delete p; *p = 1;`
Signed overflow	`INT_MAX + 1` (UB for signed; wraps for unsigned)

Returning a reference to a local

```
int& bad() {
    int local = 42;
    return local;          // BUG: local is destroyed when bad() returns
}                          // the caller gets a dangling reference

int good() {
    int local = 42;
    return local;          // fine: returns a copy of the value
}
```

> TOOLING Test with `-fsanitize=address,undefined`: it reports leaks, out-of-bounds, use-after-free and overflow with the exact line.

Tooling: warnings, sanitizers, static analysis

C++ trusts you, so let tools check what the language will not. Use them in layers, cheapest first.

LAYER	WHAT IT CATCHES	HOW
Compiler warnings	many beginner mistakes at once: uninitialized variables, signed/unsigned comparisons, missing returns	`-Wall -Wextra`, and `-Werror` in CI
Sanitizers	what only shows at run time: out-of-bounds, use after free, leaks, undefined behavior, data races	`-fsanitize=address,undefined` (and `-fsanitize=thread` for threads)
Static analyzer	rule violations, null dereferences, leaks and dead code found by reading the source, without running it	`clang-tidy`, `cppcheck`
Formatter	style, so reviews are about logic	`clang-format`

A CLEAN REPORT IS NOT A PROOF Analyzers give false positives and miss things, and a sanitizer only checks the paths your test actually ran. They reduce risk; they do not replace design, review and tests.

CODING STANDARDS Safety-critical projects (cars, medical devices) add written rules such as MISRA C++ or the AUTOSAR C++14 guidelines, checked automatically. The common themes are the ones in this course: initialize everything, no C-style casts, no naked `new` / `delete`, no silent narrowing, and no `using namespace` in headers.

Further reading by section

Where each section of this lesson is explained in more depth. Page numbers are PDF viewer pages.

How a C++ program is built

C++ Primer (Lippman, Lajoie, Moo)

- 2.6 Defining our own data structures (PDF pages 101 to 109)
- 3.1 Namespace using declarations (PDF pages 111 to 112)
- 18.2 Namespaces (PDF pages 814 to 830)

Variables and types

Programming: Principles and Practice (Stroustrup)

- 2.5 Assignment and initialization (PDF pages 49 to 52)
- 2.8 to 2.10 Type safety, conversions, type deduction with auto (PDF pages 56 to 60)

Effective Modern C++ (Meyers)

- Item 7, Distinguish () and {} when creating objects (PDF pages 67 to 76)

Operators and control flow

C++ Primer (Lippman, Lajoie, Moo)

- 4.2 to 4.3 Arithmetic, logical and relational operators (PDF pages 168 to 172)
- 4.4 to 4.5 Assignment, increment and decrement (PDF pages 173 to 178)
- 5.3 to 5.5 Conditional, iterative and jump statements (PDF pages 203 to 221)

Functions and passing arguments

C++ Primer (Lippman, Lajoie, Moo)

- 6.1 Function basics (PDF pages 231 to 236)
- 6.2 Argument passing (PDF pages 237 to 250)
- 6.3 Return types and the return statement (PDF pages 251 to 258)
- 6.4 Overloaded functions (PDF pages 259 to 264)
- 6.6 Function matching (PDF pages 271 to 275)

Programming: Principles and Practice (Stroustrup)

- 3.5 to 3.6 Functions; vector introduced (PDF pages 81 to 89)
- 7.2 to 7.3 Declarations and definitions; scope (PDF pages 194 to 202)
- 7.4 Function call and return (pass by value, by reference, by const reference) (PDF pages 203 to 218)

Pointers from scratch

C++ Primer (Lippman, Lajoie, Moo)

- 2.3 Compound types (PDF pages 79 to 87)

Programming: Principles and Practice (Stroustrup)

- 15.3 to 15.5 Memory, addresses and pointers; free store; destructors (PDF pages 452 to 463)

References

C++ Primer (Lippman, Lajoie, Moo)

- 2.3 Compound types (PDF pages 79 to 87)

Programming: Principles and Practice (Stroustrup)

- 16.1 to 16.2 Arrays, pointers and references (PDF pages 477 to 483)

const and pointers

C++ Primer (Lippman, Lajoie, Moo)

- 2.4 const qualifier (PDF pages 88 to 95)

Where variables live: stack, heap, static

Programming: Principles and Practice (Stroustrup)

- 7.2 to 7.3 Declarations and definitions; scope (PDF pages 194 to 202)

Dynamic memory: new and delete

C++ Primer (Lippman, Lajoie, Moo)

- 12.2 Dynamic arrays (PDF pages 505 to 512)

Programming: Principles and Practice (Stroustrup)

- 15.3 to 15.5 Memory, addresses and pointers; free store; destructors (PDF pages 452 to 463)

Arrays, C-strings and std::string

C++ Primer (Lippman, Lajoie, Moo)

- 3.2 Library string type (PDF pages 113 to 124)
- 3.5 Arrays (PDF pages 142 to 153)
- 3.6 Multidimensional arrays (PDF pages 154 to 161)

Programming: Principles and Practice (Stroustrup)

- 16.3 to 16.4 C-style strings; alternatives to pointer use (PDF pages 484 to 487)